

A GENERALISED EXTREME-VALUE DENSITY

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Key words: asymptotic variance-covariance matrix, extreme-value densities/variates, Gamma densities/variates, general density for logarithmic variates, logarithms of Stacy variates, parameter estimation, Stacy's generalised Gamma distribution, Weibull densities/variates.

SUMMARY

This paper presents the density for the logarithmic transformation of the generalised Gamma variate of E.W. Stacy (1962). The logarithm of any random variable of this family of variates, which includes the exponential, Chi-squared, Erlang, Gamma, Chi, Rayleigh, Maxwell, Weibull, and modular normal random variables, has proved useful in parameter estimation procedures, especially so when one notes that the arithmetic mean of the logarithms of a random sample of positive-valued variates is equivalent to the logarithm of their geometric mean. This paper discusses the statistical properties of the distribution under consideration and presents a method-of-moments technique for the joint estimation of its three parameters (location, scale, and shape). It is suggested that the method-of-moments estimators be employed as initial values in the transcendental maximum-likelihood equations, to be iterated until satisfactory convergence is attained. Asymptotic properties of these maximum-likelihood estimates are then delineated.

Parameter estimation techniques are also suggested in those cases for which two of the three parameters may be presumed known. Distributional properties of these estimators are determined in turn.

1. Introduction

An univariate probability density function, positive over the entire real axis, is defined. Indexed by three parameters (which may be interpreted as location, scale and shape parameters), the density may be shown to constitute a generalisation of the Type I limiting distribution of R.A. Fisher and L.H.C. Tippett (1928). Thus, the density is related, via logarithmic transformation, to their Type III limiting distribution, now commonly referred to as the Weibull distribution. [Cf: Gumbel (1958).]

In 1962, E.W. Stacy introduced a three-parameter generalisation of the Gamma distribution which includes as subfamilies the Weibull, Gamma, and exponential densities, among others. [Cf: Stacy and Mihram (1965).] This generalised family had been briefly

discussed somewhat earlier by A.C. Olshen (1938), who considered the distributional properties of power transformations of Gamma variates. However, we shall refer in this paper to that density as Stacy's generalised Gamma distribution, or occasionally as the Stacy density.

Since the maximum likelihood equations associated with a random sample selected from Stacy's generalised Gamma distribution reveal a dependence on the product of the observed variates, one might well speculate that the sample geometric mean would prove useful in the search for optimal parameter estimates, a result indicated as early as 1938 by N. Norris for the Pearson Type III distributions. In 1967, Z.A. Lomnicki derived the distribution for the product of independent Pearson Type III variates as well as for Weibull variates. However, the density of the product of n independent, identically distributed Stacy variates eludes precise description. Recently, H.J. Malik (1968) indicated the complexity of this fundamental form for the fundamental case ($n = 2$).

This paper describes the density for the logarithms of a large class of positive random variables, including modular normal variates. (This density is a generalisation of the extreme-value density.) Since the arithmetic mean of the sample logarithms of such variates quite naturally arises as an alternative to their geometric mean, convolutions of the present density permit probability statements regarding geometric mean estimators. Properties and characteristics of the density are delineated and, subsequent to a discussion of sufficiency and efficiency, a parameter estimation technique, motivated by maximum likelihood considerations, is outlined. Statistical properties of some of the resulting estimators are then sketched, their applicability to reliability analyses [Mihram (1965), (1972)] and to age of first marriage [Coale and McNeil (1972)] having been indicated recently.

2. Definition and Characteristics

In this paper discussion revolves about the probability density function (p.d.f.)

$$(1) f(x; a, b, c) = \{e^{b(x-a)/c}\} \{\exp[-e^{(x-a)/c}]\} / \{c \cdot \Gamma(b)\},$$

valid for all real x , for real parameters a and c (location and scale, respectively) and for positive shape parameter b , and

$$\Gamma(b) = \int_0^\infty u^{b-1} e^{-u} du,$$

denotes the standard Gamma function of Euler.

The density (1) may be shown to be closed under linear transformations $W = pX + q$; viz., W has p.d.f.

$$f(w; pa + q, b, pc).$$

With $b = 1$, the density assumes a parameterised form of the Type I limiting distribution of R.A. Fisher and L.H.C. Tippett (1928); the density (1) therefore constitutes a generalisation thereof. Furthermore, with $b = 1$, the density of $U = \exp(X)$ represents one of the limiting forms (Type III) of Fisher and Tippett (1928) commonly denominated the Weibull distribution (1951) whenever $d = 1/c > 0$.

The Laplace transform of the density (1) is given by

$$(2) \quad m_X(t) = E[e^{tX}] = e^{at} \Gamma(b + ct) / \Gamma(b),$$

provided that $(b + ct) > 0$.

Thus, the mean of the density (1) is given by

$$(3) \quad E(X) = m'_X(0) = a + c\psi(b),$$

where $\psi(v) = d \ln \Gamma(v)/dv$, the digamma function, as tabulated by M. Abramowitz and I.A. Stegun (1964). Successive differentiations of the moment-generating function associated with $Z = (X - a)/c$ provide the central moments:

$$(4) \quad \mu_2(Z) = \text{Var}(Z) = c^{-2} \text{Var}(X) = \psi'(b),$$

and

$$(5) \quad \mu_3(Z) = c^{-3} \mu_3(X) = \psi''(b),$$

yet

$$(6) \quad \mu_4(Z) = c^{-4} \mu_4(X) = \psi'''(b) + 3\psi(b),$$

where the acute accents here denote differentiation with respect to the argument of the derived function. [Cf: Stacy and Mihram (1965).]

In passing, we note that the ratio of moments

$$(7) \quad \frac{[\mu_3(Z)]^2}{[\mu_2(Z)]^3} = \frac{[\mu_3(X)]^2}{[\mu_2(X)]^3} = \frac{[\psi''(b)]^2}{[\psi'(b)]^3}$$

is independent both of the scale and of the location parameter.

Equating the derivative of function (1) to zero and solving therefore yields the

mode of the density:

$$(8) \quad \text{mode}(X) = a + c \ln b.$$

Thus, reference to Equation (3) reveals that

$$(9) \quad \text{mode}(X) - E(X) = c[\ln b - \psi(b)].$$

Since, as is shown by Edwards (1930, vol. II),

$$\ln b > \psi(b), \text{ for all } b > 0,$$

yet $\psi(b) \rightarrow \ln b$ as $b \rightarrow \infty$,

we see that the modal value precedes (follows) the mean value whenever $c < 0$ (> 0) and that the two approach one another as b increases. As a matter of fact, since

$$f(x; a, b, -c) = f(-x; -a, b, c) \text{ for all } x,$$

the absolute difference between the mode and the median is symmetric in c (for fixed b).

The cumulative distribution function for the variate X is

$$(10) \quad F(x; a, b, c) = \begin{cases} \gamma(e^{(x-a)/c}; b), & \text{for } c > 0 \\ 1 - \gamma(e^{(x-a)/c}; b), & \text{for } c < 0 \end{cases}$$

where $\gamma(t; b) = \int_0^t u^{b-1} e^{-u} du / \Gamma(b)$,

the incomplete Gamma ratio, as tabulated by Karl Pearson (1934). Consequently, expressions for the quantiles, such as the median, involve in general the solution of integral equations.

Noteworthy, however, is the proportion of the density to the left of its modal value; from Equations (8) and (10),

$$(11) \quad P[X \leq \text{mode}(X)] = \begin{cases} \gamma(b; b), & \text{for } c > 0 \\ 1 - \gamma(b; b), & \text{for } c < 0. \end{cases}$$

Reference to the aforementioned tabulations of Pearson (1934) indicates that this proportion increases (decreases) to $\frac{1}{2}$ as b increases, whenever $c < 0$ ($c > 0$). Due to the

limiting relationship of the mode and the mean, this same result obtains for the proportion of the density to the left of the mean. Thus, as b becomes infinitely great, the mean, the mode, and the median become relatively nearer. —

The rate at which the mean moves to the right with increasing shape parameter is disclosed via reflections on the means of standardized variates, $Z = (X-a)/c$, having p.d.f. $f(z; 0, b, 1)$, and another Z' , having p.d.f. $f(z'; 0, b+1, 1)$. The respective means are $\psi(b)$ and $\psi(b+1)$, their difference, $\psi(b+1) - \psi(b) = b^{-1}$, being the derivative of $\ln b = \ln[\Gamma(b+1)/\Gamma(b)]$. Thus, for standardized variates, a unit increase in the shape parameter shifts the mean by an amount inversely proportional to the shape parameter value from which the shift is made. More generally, if X has the p.d.f. $f(x; a, b, c)$ and X is distributed according to $f(x; a, b+1, c)$, then

$$(12) \quad E(X') - E(X) = c[\psi(b+1) - \psi(b)] = c/b.$$

3. Sufficiency and Minimum Variance Considerations

Let $x_1, x_2, \dots, x_n$ denote the elements of a complete random sample of size n from density (1). The log-likelihood equations become:

$$(13) \quad \begin{cases} \frac{\partial \ln L}{\partial a} = \frac{-nb}{c} + \frac{1}{c} \cdot \left\{ \sum_{i=1}^n e^{(x_i - a)/c} \right\}, \\ \frac{\partial \ln L}{\partial b} = \sum_{i=1}^n (x_i - a)/c - n \psi(b), \\ \text{and} \\ \frac{\partial \ln L}{\partial c} = \frac{-b \cdot \sum_{i=1}^n (x_i - a)}{c^2} + c^{-2} \sum_{i=1}^n (x_i - a) \cdot e^{(x_i - a)/c} \cdot \frac{n}{c} \end{cases}$$

Since $Z = (X - a)/c$ has p.d.f. $f(z; 0, b, 1)$, one can show that $E[Z^s \exp(tZ)] = b E[W^s]$, where W has the density $f(w; 0, b+1, 1)$ and $E[Z^s \exp(tZ)] = \Gamma(b+t) E(V^s)/\Gamma(b)$, where V is a random variable with density $f(v; 0, b+t, 1)$.

The inverse, $A = V^{-1}$, of the symmetric asymptotic variance-covariance matrix for the minimum likelihood estimates, $(\hat{a}, \hat{b}, \hat{c}) \equiv (\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3)$, has elements:

$$(14) \quad a_{11} = nb/c^2,$$

$$(15) \quad a_{12} = a_{21} = n/c,$$

$$(16) \quad a_{13} = a_{31} = nb \psi(b+1)/c^2,$$

$$(17) \quad a_{22} = n \cdot \psi' (b),$$

$$(18) \quad a_{23} = a_{32} = n \psi(b)/c, \text{ and}$$

$$(19) \quad a_{33} = n \{1 + b [\psi' (b + 1) + \psi^2 (b + 1)] \} / c^2.$$

Thus, given two of the three parameters' values, the minimum attainable variances for the unbiased estimates, $a^*(B, C)$, $b^*(A, C)$, and $c^*(A, B)$, of the respective parameters become [Cf: C. R. Rao (1945)]:

$$(20) \quad \begin{cases} \text{Var} (a^*) \geq 1/E \left[\frac{\partial \ln L}{\partial a} \right]^2 = C^2/nB, \\ \text{Var} (b^*) \geq 1/E \left[\frac{\partial \ln L}{\partial b} \right]^2 = 1/n \psi' (b), \\ \text{and similarly} \\ \text{Var} (c^*) \geq c^2 / n \{ 1 + B [\psi' (B + 1) + \psi^2 (B + 1)] \}. \end{cases}$$

Nonetheless, the form of the density (1) does not conform to that required by E.J.G. Pitman's factorization criterion (1936). Thus, a set of minimal sufficient statistics does not exist for the parameters a , b , and c . As a result, the variance bounds of Equations (20) shall not likely be attained in any joint parameter estimation procedure (except as an asymptotic property for the estimators).

4. Parameter Estimation

Simultaneous solution of the Equations (13), each equated to zero, provides the maximum likelihood estimators ($\hat{a}$, $\hat{b}$, and $\hat{c}$), asymptotically normal and consistent. Solutions to the transcendental equations are not available in closed form, so that iterative techniques, probably best performed via programmed electronic computers, are recommended. Initial values for b_0 and c_0 for such a procedure may be obtained in the manner explained by Stacy and Mihram (1965).

Iterations through the Equations (13) may now proceed by substitution of b_0 and c_0 into, say, the first to obtain the initial location parameter estimate

$$a_1 = c_0 \ln \left(\sum_{i=1}^n e^{x_i/c_0} / n b_0 \right).$$

Iterations may then proceed through the equations (13) until satisfactory convergence is attained. As a matter of fact, if using the Newton-Raphson method, one may verify that the convergence conditions required by Kale (1962) are satisfied. The asymptotic

variances of the resulting estimates are found from the covariance matrix [See the Equations (14) through (19)].

In many applications, knowledge of one or more parametric values may preclude the necessity for such cumbersome estimation techniques. For example, with the shape and scale parameters equal respectively to the constants B and C , Equation (3) suggests the unbiased location parameter estimate:

$$(21) \quad a^*_1(B, C) = \bar{x} - C \cdot \psi(B),$$

where $\bar{x}$ represents the sample mean. This estimator has its variance given by

$$\text{Var} [a^*_1(B, C)] = \text{Var}(X)/n = C^2 \psi'(B)/n > C^2/(n \cdot B).$$

Though its distribution remains concealed, its moment-generating function is available via that of $\bar{x}$:

$$(22) \quad \begin{aligned} m(t) &= e^{-tC\psi(B)} m_{\bar{x}}(t) \\ &= e^{-tC\psi(B) + at} \Gamma^n(B + Ct/n) / \Gamma^n(B). \end{aligned}$$

An alternative unbiased estimate of the location parameter may be constructed. For the case where $b = B$ and $c = C$ (constants), its variance more nearly attains the minimum variance bound of the first of the Equations (20). Since $U = e^X$ has Stacy's generalised Gamma distribution (1962), it follows from its properties [Cf: Stacy and Mihram (1965)] and the additive reproductivity of the two-parameter Gamma distribution, that $T_n = \ln [\sum_{i=1}^n \exp(X_i/C)]^C$ has the p.d.f. $f(t; a, nB, C)$, so that

$$(23) \quad a^*_2(B, C) = T_n - C\psi(nB) = \ln \left[\sum_{i=1}^n e^{X_i/C} / e^{\psi(nB)} \right]^C,$$

having the p.d.f. $f(a_2; a - C\psi(nB), nB, C)$, becomes an unbiased estimate for the location parameter with variance

$$(24) \quad \text{Var}[a^*_2(B, C)] = C^2 \psi'(nB) > C^2/nB, \text{ yet } < C^2 \psi'(B)/n.$$

In the event that the two parameters a and c are presumed known as the respective constants A and C , an unbiased estimate of the shape parameter results from

$$(25) \quad b^*(A, C) = n^{-1} \sum_{i=1}^n e^{(x_i - A)/C},$$

the statistic having one of the generalised Gamma distributions of Stacy with variance

$$(26) \text{Var}[b^*(A, C)] = b/n > 1/n \psi'(b).$$

An estimator with optimal variance is provided by

$$(27) \psi^*(A, C) = \sum_{i=1}^n (x_i - A)/nC,$$

unbiased for $\psi(b)$ with variance

$$(28) \text{Var}[\psi^*(A, C)] = \psi'(b)/n,$$

the minimum attainable variance for estimates which are unbiased for $\psi(b)$.

Finally, in the event that $a = A$ and $b = B$, leaving the scale parameter c unknown, an unbiased estimate may be obtained as

$$(29) c^*(A, B) = \sum_{i=1}^n (x_i - A)/n\psi(B),$$

with variance

$$(30) \text{Var}[c^*(A, B)] = c^2 \psi'(B)/n\psi^2(B).$$

Nonetheless, an unbiased estimator, with optimal variance properties, for this parameter remains elusive at this writing.

5. Summary and Conclusions

A three-parameter probability density function has been defined, its properties delineated, and a technique for joint parameter estimation outlined in the event that a complete random sample is available. Asymptotic properties of the resulting estimates have been described, though precise non-asymptotic distributional properties remain concealed.

Finally, since the density serves to describe the distribution of logarithmic transformations of numerous positive random variables (e.g., Gamma, Weibull, Chi, Chi-squared, exponential, Rayleigh, and modular normal), unbiased parameter estimates have been provided whenever any two of the three parameters are known. In these cases, the variances (and, where possible, the distributions) of these estimates have been indicated.

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